

HelixVPN-Interaction-Specs

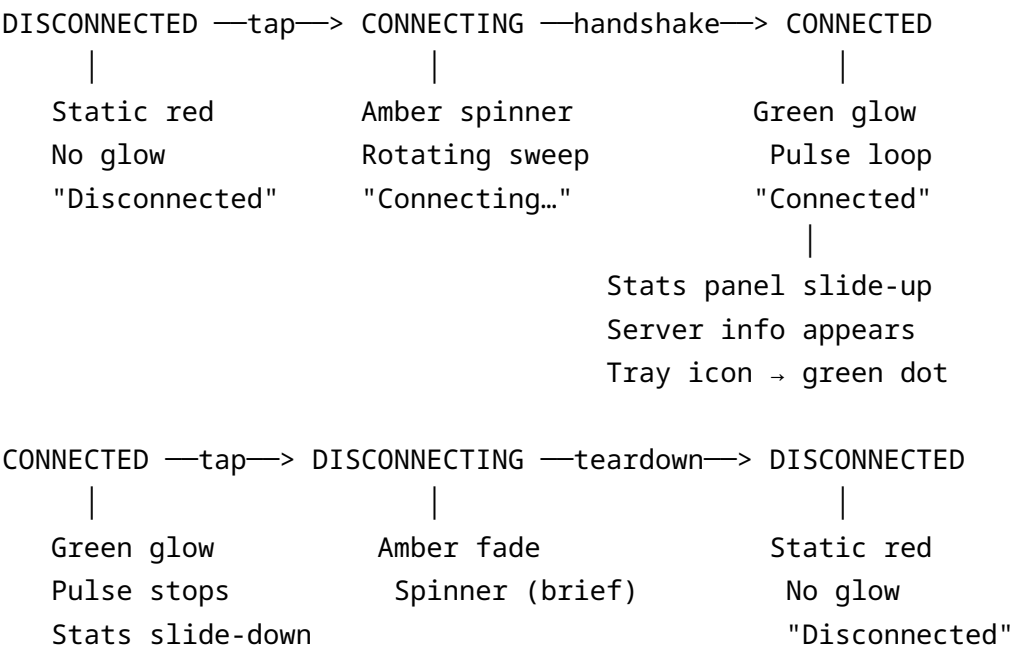
Helix VPN — Interaction Patterns & Animation Specs

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1. Connection Flow (Critical Path)

The connection lifecycle is the most important interaction in the product. Every animation serves to communicate state clearly and build trust.

Flow Diagram



Timing Table

Phase	Event	Duration	Easing	Visual
1a	Button press	100ms	ease-out	Scale 0.95, no animation on release
1b	Disconnected → Connecting	200ms	ease-in-out	Ring color red → amber crossfade
1c	Spinner appears	300ms	ease-out	Fade in + rotation start
1d	“Connecting...” text	200ms	ease-out	Fade in
2a	Handshake progress	500–2500ms	—	Continuous spinner (360°/1500ms)
2b	Optional progress ring	—	—	Arc sweep around button
2c	Error during handshake	—	800ms flash	Red fast pulse
3a	Connecting → Connected	300ms	ease-in-out	Ring amber → green crossfade
3b	Spinner → Checkmark	400ms	spring(1,80,10)	Spinner fades, checkmark scales 0 → 1 (bounce)
3c	Glow establishes	500ms	ease-out	20px blur, 30% opacity, pulse loop begins
3d	Stats panel slide-up	400ms	cubic-bezier(0.4,0,0.2,1)	translateY from below + opacity 0 → 1
3e	Status text update	200ms	ease-out	“Connected” + server info
3f	Tray icon update	150ms	—	Badge turns green

Disconnect: Reverse sequence, 800ms total: - Green → red ring (300ms)
- Stats panel slide-down (300ms) - Glow fades (200ms) - Tray icon → outline/disconnected

Animation Code (CSS Keyframes)

```
@keyframes connect-pulse {
  0%, 100% { box-shadow: 0 0 20px rgba(76, 175, 80, 0.3); }
  50%      { box-shadow: 0 0 30px rgba(76, 175, 80, 0.5); }
}

@keyframes spin {
  from { transform: rotate(0deg); }
  to   { transform: rotate(360deg); }
}

@keyframes glow-pulse {
  0%, 100% { opacity: 0.2; }
  50%      { opacity: 0.4; }
}

@keyframes ripple {
  0%   { transform: scale(0); opacity: 0.5; }
  100% { transform: scale(4); opacity: 0; }
}
```

2. Page Transitions

Platform	Transition	Duration	Easing	Notes
Desktop	Crossfade	200ms	ease-in-out	Between pages within window
Desktop modal	Fade in backdrop	200ms	ease-out	Backdrop opacity 0 → scrim
Desktop modal	Content scale in	300ms	spring	Content scale 0.95 → 1 + fade
		300ms	ease-out	

Platform	Transition	Duration	Easing	Notes
Desktop drawer	Slide from right			300px width, scrim behind
Desktop settings tab	Crossfade	200ms	ease-in-out	Tab content only
Android	Push (slide right)	300ms	ease-in-out	Enter: translateX 100% → 0, Exit: 0 → -30%
Android	Bottom sheet	300ms	ease-out	translateY 100% → 0
iOS	Push (Cupertino)	350ms	ease-in-out	Standard Cupertino transition
iOS	Modal cover	400ms	ease-out	translateY from bottom
iOS	Bottom sheet	350ms	ease-out (iOS 15+)	Interactive drag dismissal
HarmonyOS	Smooth spring	350ms	ArkUI spring	Native ArkUI physics
Aurora	Slide with depth	Silica default	Silica default	Horizontal parallax
Web	Instant	0ms	—	No transitions (popup is small)

3. Micro-Interactions

Interaction	Animation	Duration	Easing	Trigger
Button hover		150ms	ease-out	

Interaction	Animation	Duration	Easing	Trigger
	Scale 1.02, lift 1px			Mouse enter
Button press	Scale 0.98	100ms	ease-out	Mouse down
Button release	Scale 1.0	100ms	ease-out	Mouse up
Toggle/Switch on	Thumb translateX + bg color	200ms	ease-in-out	Tap
Toggle/Switch off	Thumb translateX reverse	200ms	ease-in-out	Tap
Checkbox check	Checkmark stroke draw	200ms	ease-out	Tap
Card hover	translateY -2px, shadow inc	200ms	ease-out	Mouse enter
List item tap	Ripple from touch point	400ms	ease-out	Tap (mobile)
Pull-to-refresh	Rotation + arc sweep	1000ms	linear	Pull past threshold
Toast appear	Slide in top-right + fade	300ms	ease-out	Trigger
Toast dismiss	Slide out + fade	300ms	ease-in	4000ms timer
Copy feedback	Checkmark flash	500ms	ease-out	Tap copy
Focus ring	Outline fade in	200ms	ease-out	Keyboard Tab
Scroll bar	Fade in on scroll	200ms	ease-out	Scroll start
Scroll bar	Fade out	500ms	ease-in	Scroll end + 1s delay
Icon morph	Fill → Outline / Chevron → X	200ms	ease-in-out	State change
Skeleton shimmer	Gradient sweep	1200ms	linear	Data loading >300ms
Progress bar		300ms	ease-out	

Interaction	Animation	Duration	Easing	Trigger
	Smooth width			Upload/download
Server item selected	Tint bg fade in	200ms	ease-out	Selection
Notification badge	Scale bounce 0 → 1	300ms	spring	New notification

Reduced Motion Override

```
@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after {
    animation-duration: 0.01ms !important;
    animation-iteration-count: 1 !important;
    transition-duration: 0.01ms !important;
  }

  .connect-toggle {
    /* Static glow – no pulse */
    animation: none;
  }

  .connect-toggle.connected {
    box-shadow: 0 0 15px rgba(76, 175, 80, 0.4); /* Fixed glow */
  }

  .connect-toggle.connecting {
    /* Static spinner icon, no rotation */
    /* Replace with static "uploading" icon */
  }
}
```

Fallback behaviors: - Pulse → static glow - Page transitions → instant (no animation) - Spinners → static indicator + “Loading...” text - Ripples → instant opacity change - Graphs → static (no draw animation) - Toast → instant appear

4. Gesture & Navigation Patterns

Desktop Keyboard Navigation

Key	Action
Tab	Move focus forward
Shift+Tab	Move focus backward
Enter/Space	Activate focused element
Escape	Close modal/dialog/menu
Ctrl/Cmd+K	Quick connect/disconnect
Ctrl/Cmd+Shift+S	Open server selection
Ctrl/Cmd+,	Open preferences
Ctrl/Cmd+Q	Quit application

Focus ring: 2px --hx-border-focus outline, 2px offset. Visible only on keyboard navigation (not mouse click).

Mobile Gestures

Gesture	Platform	Action
Swipe left (server item)	Android	Delete favorite (72dp action)
Swipe right (server item)	Android	Quick connect (72dp, green bg)
Swipe left (setting item)	Android	Reset to default
Pull down (server list)	All mobile	Refresh server list
Pull down (main screen)	Aurora	Open pulley menu
Long-press (server)	All mobile	Context menu: copy IP, details
Long-press (launcher icon)	Android	App shortcuts: connect, servers
Back swipe (left edge)	iOS/Android	Navigate back
Bottom sheet drag	All mobile	

Gesture	Platform	Action
		Sheet dismiss (velocity >300dp/s)

Haptic Feedback

Interaction	Platform	Type
Connect tap	iOS	Heavy impact
Connect success	iOS	Success notification
Connect failure	iOS	Error notification
Swipe action	iOS	Medium impact
Long-press trigger	iOS	Medium impact
Pull-to-refresh threshold	iOS	Light impact

5. Loading States

Timing-Based Display

Time	Loading Indicator	Behavior
0–300ms	None (instant response)	Content already displayed
300ms–2s	Skeleton loading	Shimmer animation on placeholder shapes
>2s	Spinner + status text	Indeterminate progress + “Still connecting...”

Skeleton Screen Specification

- Shape: Rounded rectangles matching content layout
- Color: bg-tertiary (light) / bg-secondary (dark)
- Animation: Shimmer gradient sweep (1200ms linear, diagonal)
- Trigger: Data fetch >300ms

Spinner Specifications

Size	Size (px)	Stroke	Usage
Small	24×24	3px	Inline loading, button replacement
Medium	40×40	3px	Section loading
Large	56×56	4px	Full-page loading

Color: --hx-primary-500 (track: surface variant, 20% opacity) Duration: 1000ms per rotation

Progress Bar

- Height: 4px (linear, top of content area)
- Fill: Primary color gradient
- Track: Surface variant
- Animation: Smooth width transition, 300ms
- Determinate: Percentage-based width
- Indeterminate: Continuous sweeping motion

6. Notification Patterns

Desktop Notifications

Event	Type	Action
Connected	Toast (top-right)	—
Disconnected	Toast + tray icon highlight	“Tap to reconnect” (tray)
Connection error	Toast (error style)	“Tap to retry”
Reconnecting	Toast (info style)	—
Kill Switch Active	Critical alert	“Traffic blocked — no leakage”
Update available	Toast (info)	Auto-download start
Auto-connect	Silent	No notification

Mobile Notifications

Platform	Type	Content
Android	Foreground service notification	Status: Connected to US East, Data: 1.2 GB, Disconnect button
iOS	System notification (non-intrusive)	Status updates via NE provider
HarmonyOS	Notification via HMS Push Kit	Status, data usage

Notification Design

[icon] Connected to US East IP: 203.0.113.45 (green) [×]	← Toast: 48px height 4px left border
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7. Screen Reader Support

ARIA / Accessibility Attributes

Element	Attribute	Value
Connection toggle	aria-label	“Toggle VPN connection — currently connected”
Server list items	aria-label	“US East server, 23 milliseconds latency”
Status text	aria-live	“polite”
Connection stats	aria-live	“polite”
Modal	role	“dialog”
Modal close	aria-label	“Close dialog”

Element	Attribute	Value
Navigation	role	“navigation”
Main content	role	“main”
Alerts	role	“alert”
Progress	role	“progressbar”

Connection State Announcements

State	Announcement
Connected	“Connected to US East, 23 milliseconds latency”
Connecting	“Connecting to US East server”
Disconnected	“Disconnected — network traffic is not protected”
Kill Switch	“Kill switch active — all traffic blocked”
Error	“Connection failed — unable to reach server”

8. Platform-Specific Pattern Adaptations

macOS

- Standard traffic light window controls — never custom chrome
- Native menu bar (HelixVPN > About, Preferences, Services, Quit)
- Cmd+ shortcuts for all primary actions
- Vibrancy effect on sidebar
- SF Pro font throughout

Windows 11

- Mica material on title bar and background surfaces
- NavigationView sidebar for settings (WinUI-style)
- Snap layout support (window zones)
- Context menu styling (system-native)

Linux (GNOME)

- Header bar integrated with CSD when available
- Flat design — no unnecessary shadows
- GTK theme detection for color adaptation

- Compact/comfortable density configurable

Android

- Material 3 with dynamic color (wallpaper-derived primary on 12+)
- Bottom navigation bar (3 items: Home, Servers, Settings)
- Edge-to-edge display behind system bars
- Quick Settings tile (1×1, monochrome)

iOS

- Cupertino navigation bar with blur
- Inset grouped table sections for settings
- System blur materials (thin, regular, thick)
- SF Symbols for all icons
- Standard iOS back gesture + swipe navigation

HarmonyOS

- ArkUI design language with smooth spring physics
- HarmonyOS Sans typography
- Service widget (form card) for home screen
- Distributed capability framework (future cross-device)

Aurora OS

- Silica gesture navigation (no bottom nav)
- Pulley menu for primary actions
- Cover page for background status
- Ambiance-aware coloring
- Sailfish Silica component library (no Qt Quick Controls)